

Preamble

A central ingredient in field theory is diagrammatics. In classical φ^4 theory, there are, to one loop, only two diagrams that determine the critical exponents:



I will briefly review how they determine the exponents below. I can see that they are essentially a matter of counting contributions. However:

- it's not about their *asymptotic* combinatorics, as far as I can tell;
- std. DP problems are often *not* of this form. There, it's usually about summing over *all* diagrams, or many, in a systematic way.

1 Standard φ^4 is not about asymptotic numbers of diagrams

(a) Tadpole

$$\begin{aligned}
 \text{tadpole} &\hat{=} \int \frac{d^d k}{(k^2 + r)} = K_d \int_0^\infty dk \frac{k^{d-1}}{k^2 + r} \\
 &= K_d \frac{1}{2} \int du \frac{u^{d/2-1}}{u + r} = K_d \frac{1}{2} r^{d/2-1} \frac{\Gamma(1 - \frac{d}{2}) \Gamma(1)}{\Gamma(\frac{d}{2})} \\
 &= A r r^{-\varepsilon/2} \boxed{\frac{\Gamma(\varepsilon/2)}{1-d/2}}
 \end{aligned}$$

with $A = \frac{1}{2} K_d$, $\Gamma(1) = 1 = \Gamma(2)$, $\Gamma(1 - \frac{d}{2})(1 - \frac{d}{2}) = \Gamma(2 - \frac{d}{2})$, $\varepsilon = 4 - d$.

Combinatorial factor: $4 \cdot 3 = \underbrace{\frac{4 \cdot 3}{2}}_{\text{loop}} \cdot \underbrace{2}_{\text{external}}$

(b) Bubble (sunset / two-line)

$$\begin{aligned}
 \text{bubble} &\hat{=} \int \frac{d^d k}{(k^2 + r)^2} = K_d \frac{1}{2} \int du \frac{u^{d/2-1}}{(u + r)^2} \\
 &= K_d \frac{1}{2} r^{d/2-2} \frac{\Gamma(2 - \frac{d}{2})}{\Gamma(2) \Gamma(\frac{d}{2})} = A r^{-\varepsilon/2} \boxed{\Gamma(\varepsilon/2)}
 \end{aligned}$$

Combinatorial factor: $\binom{4}{2} \binom{4}{2} \cdot 2 \cdot 4! \cdot \frac{1}{2}$ (legs to form loop · external · spinlet relabel)

So

$$g = 2A r^{-\varepsilon/2} u_4 \quad (\text{sd. nonlinearity}), \quad \Gamma(\varepsilon/2) = \frac{2}{\varepsilon} + \text{h.o.t.}$$

Resulting flow and exponent

$$(\beta) \quad \beta_u = -\varepsilon \cdot 4! g + 36 \cdot 4! g^2 \xrightarrow{\beta_u=0} g^* = \frac{\varepsilon}{36}$$

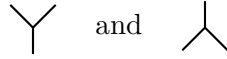
$$(\gamma) \quad Z_r = 1 + 12g \Rightarrow \gamma_r = 12g^* = \frac{1}{3}\varepsilon$$

$$(\nu) \quad \frac{1}{\nu} = 2 - \gamma_r \Rightarrow \nu = \frac{1}{2} \frac{1}{1 - \gamma_r/2} = \frac{1}{2} + \frac{\gamma_r}{4} + \dots = \frac{1}{2} + \frac{1}{12}\varepsilon$$

The exponent is determined by the **ratio** of the integrals and the combinatorial factors as highlighted. This is **not** about asymptotics, just a matter of counting.

2 My current favourite DP problem

Consider the two vertices



and label them as H -vertices with leg occupations $(m_1, m_2 | n_1, n_2)$:



Their couplings are given by Kronecker structure

$$\left[\begin{array}{c} m_1 \ n_1 \\ \text{Y-shape} \\ m_2 \ n_2 \end{array} \right] = \delta_{m_2, n_2} \delta_{m_1-1, n_1+1} + \delta_{m_2, n_2} n_1 \delta_{m_1-1, n_1-1} \\ - \delta_{m_2, n_2+1} \delta_{m_1-1, n_1+1} - \delta_{m_2, n_2-1} n_2 \delta_{m_1-1, n_1}.$$

One-loop (warm-up).

$$\begin{array}{c} 2 \quad 1 \\ \text{---} \quad \text{---} \\ 1 \quad 1 \\ | \quad | \\ 0 \quad 0 \end{array} = \frac{1}{-\hat{i}\omega + r + k_o} \frac{1}{-\hat{i}\omega + r + k_o} \nu \begin{array}{c} 1 \quad 1 \\ \text{Y-shape} \\ 0 \quad 0 \end{array}$$

Two-loop.

$$\begin{array}{c} \text{---} \quad \text{---} \quad \text{---} \quad \text{---} \quad \text{---} \\ | \quad | \quad | \quad | \quad | \\ \text{---} \quad \text{---} \quad \text{---} \quad \text{---} \quad \text{---} \end{array} = \frac{1}{-\hat{i}\omega + r + k_o} \frac{1}{-\hat{i}\omega + r + k_o} \sum_{\ell_1, \ell_2} \nu \begin{array}{c} 1 \quad \ell_1 \\ \text{Y-shape} \\ 0 \quad \ell_2 \end{array} \nu \left(\begin{array}{c} \ell_1 \quad 1 \\ \text{Y-shape} \\ \ell_2 \quad 0 \end{array} + \begin{array}{c} \ell_2 \quad 0 \\ \text{Y-shape} \\ \ell_1 \quad 1 \end{array} \right) \\ \times \underbrace{\int d\omega' \frac{1}{-\hat{i}(\omega - \omega') + r + \ell_1 k_o} \frac{1}{-\hat{i}\omega' + r + \ell_2 k_o}}_{\frac{1}{-\hat{i}\omega + 2r + (\ell_1 + \ell_2)k_o}}$$

By Mathematica:

$$= \frac{1}{-\hat{i}\omega + r + k_o} \frac{1}{-\hat{i}\omega + r + k_o} \nu^2 \frac{2}{-\hat{i}\omega + 2r + k_o}.$$

Three-loop.

$$\begin{aligned}
\text{Diagram} &= \frac{1}{-\hat{i}\omega + r + k_o} \frac{1}{-\hat{i}\omega + r + k_o} \sum_{\ell_1 \dots \ell_4} \nu \begin{array}{c} \ell_1 \ell_1 \\ \diagup \quad \diagdown \\ 0 \quad \ell_2 \end{array} \nu \\
&\times \left(\begin{array}{c} \ell_1 \ell_3 \\ \diagup \quad \diagdown \\ \ell_2 \ell_4 \end{array} + \begin{array}{c} \ell_2 \ell_4 \\ \diagup \quad \diagdown \\ \ell_1 \ell_3 \end{array} \right) \left(\begin{array}{c} \ell_3 1 \\ \diagup \quad \diagdown \\ \ell_4 0 \end{array} + \begin{array}{c} \ell_4 0 \\ \diagup \quad \diagdown \\ \ell_3 1 \end{array} \right) \\
&\underbrace{\hspace{10em}}_{\frac{1}{-\hat{i}\omega + 2r + (\ell_1 + \ell_2)k_o}} \underbrace{\hspace{10em}}_{\frac{1}{-\hat{i}\omega + 2r + (\ell_3 + \ell_4)k_o}}
\end{aligned}$$

By Mathematica:

$$= \frac{1}{-\hat{i}\omega + r + k_o} \frac{1}{-\hat{i}\omega + r + k_o} \nu^3 \frac{4}{(-\hat{i}\omega + 2r + k_o)^2}. \quad \text{Hm.}$$

General n -loop sum. So we need

$$\begin{aligned}
&\sum_{\ell_1, \dots, \ell_{2n}} \begin{array}{c} 1 \ell_1 \\ \diagup \quad \diagdown \\ 0 \quad \ell_2 \end{array} \cdot \left(\begin{array}{c} \ell_1 \ell_3 \\ \diagup \quad \diagdown \\ \ell_2 \ell_4 \end{array} + \begin{array}{c} \ell_2 \ell_4 \\ \diagup \quad \diagdown \\ \ell_1 \ell_3 \end{array} \right) \left(\begin{array}{c} \ell_3 \ell_5 \\ \diagup \quad \diagdown \\ \ell_4 \ell_6 \end{array} + \begin{array}{c} \ell_4 \ell_6 \\ \diagup \quad \diagdown \\ \ell_3 \ell_5 \end{array} \right) \cdots \\
&\left(\begin{array}{c} \ell_{2n} \ell_{2n-1} \\ \diagup \quad \diagdown \\ \ell_{2n} \ell_{2n} \end{array} + \begin{array}{c} \ell_{2n} \ell_{2n} \\ \diagup \quad \diagdown \\ \ell_{2n} \ell_{2n-1} \end{array} \right) \left(\begin{array}{c} \ell_{2n-1} 1 \\ \diagup \quad \diagdown \\ \ell_{2n} 0 \end{array} + \begin{array}{c} \ell_{2n} 0 \\ \diagup \quad \diagdown \\ \ell_{2n-1} 1 \end{array} \right) = 2^n.
\end{aligned}$$

(I can compute that using a very different route.)

How can I see that?
Generating functions? Really?

Response / Commentary

Reading Gunnar's two questions

Gunnar's note (and his follow-up email) leaves us with two concrete charges. We have regrouped them around his sharpest sentence — “*What is $T(z)$ and how does Claude calculate it?*” — because that is the place his draft and our draft were talking past each other.

Q1 (technical, §2). “How does Claude reduce my counting problem to a 2×2 matrix? When I do this I either end up with an infinite matrix or a generating function in four variables, so it's a mess.” This is the only thing he actually wants from us right now. *Everything below in R2 is for him.*

Q2 (philosophical, §1). Standard φ^4 exponents come from a finite Wick ratio ($4 \cdot 3$ vs. $\binom{4}{2}^2 \cdot 2 \cdot 4!/2$), not from $n!^\alpha$ asymptotics. So the AC framing of the Gribov draft was advertising a route the φ^4 exponents don't need. *R1 below addresses this — it is short.*

R1. On φ^4 counting — AC is the counting machine

He is right that the φ^4 exponents come from *counting*, not from $n!^\alpha$ asymptotics. “Counting” and “asymptotic combinatorics” are not different machines — they are the same generating function evaluated at different orders:

- *Low-order coefficients* of the labelled-diagram generating function $\mathcal{Z}(g, r)$ are the Wick weights $4 \cdot 3$ (tadpole) and $\binom{4}{2}^2 \cdot 2 \cdot 4! / 2$ (bubble) — read off as Gaussian-moment derivatives, no hand contractions (Appendix A).
- *Asymptotic growth* of the same coefficients (radius of convergence, $n!^\alpha$ prefactor) controls the Tier-B observables (LERW 5/4, TAP 1.624, KPZ replica), where no β -function fixed point is available.

So AC contains φ^4 exponent-extraction as the leading-order truncation of its own generating function. The “asymptotic” label in our Gribov draft was poorly chosen; we should rewrite the framing as “ φ^4 exponents are the low-order residue of the same generating function whose $n \rightarrow \infty$ tail gives Tier-B.” That reframing is what makes the answer to Q1 below not look like a coincidence.

R2. AC predicts the 2^n from three structural facts

His complaint is correct as posed: the four-term coupling

$$V(m_1, m_2; n_1, n_2) = \underbrace{\delta_{m_2, n_2} \delta_{m_1-1, n_1+1}}_{T1} + \underbrace{\delta_{m_2, n_2} n_1 \delta_{m_1-1, n_1-1}}_{T2} - \underbrace{\delta_{m_2, n_2+1} \delta_{m_1-1, n_1+1}}_{T3} - \underbrace{\delta_{m_2, n_2-1} n_2 \delta_{m_1-1, n_1}}_{T4}$$

gives an infinite-dimensional operator on Fock space $\mathcal{F} = \text{span}\{|m_1, m_2\rangle\}$, and the corresponding generating polynomial $\hat{V}(z_1, z_2; w_1, w_2)$ has unbounded degree. The reduction is not a property of $\hat{V}$ itself — it is forced by the boundary $|(1, 0)\rangle$. Three structural inputs predict the eigenvalue 2 before any matrix arithmetic; the polynomial collapse $\hat{T} = (z_1 - z_2)(w_1 - w_2)$ then exhibits it in one line.

Structural input 1: Conservation by boundary $\Rightarrow$ sector dimension. The four terms shift total particle number $\Delta N \equiv (m_1 + m_2) - (n_1 + n_2)$ by $\{2, 0, 3, 0\}$. So V is *not* particle-conserving on $\mathcal{F}$. But on inputs with $m_1 + m_2 = 1$, T1 and T3 require $m_1 \geq 2$ and vanish identically; T2 and T4 conserve $\Delta N = 0$. So when the boundary fixes $|(1, 0)\rangle$ (one particle), every internal state in the chain stays in the single-particle sector. *This sector has dimension 2 — predicted, not computed — because $\#\{(m_1, m_2) \in \mathbb{Z}_{\geq 0}^2 : m_1 + m_2 = 1\} = 2$.*

Structural input 2: Term 2 is a number operator $\Rightarrow$ trace. T2 is the only diagonal-preserving term ($m_1 = n_1, m_2 = n_2$), and its weight on the diagonal is $n_1 = m_1$. So $V_{\text{diag}}(m_1, m_2) = m_1$ identically — T2 acts as the rail-1 number operator. The rail-flipped vertex $\tilde{V}$ ($\tilde{V}(m_1, m_2; n_1, n_2) \equiv V(m_2, m_1; n_2, n_1)$) is rail-2’s number operator on the diagonal: $\tilde{V}_{\text{diag}} = m_2$.

$$\boxed{T_{\text{diag}}(m_1, m_2) = V_{\text{diag}} + \tilde{V}_{\text{diag}} = m_1 + m_2 = N \quad (\text{total particle number}).}$$

On the N -particle sector, every diagonal entry of T equals N , so

$$\text{Tr}(T) = N \cdot \dim(N\text{-sector}).$$

For $N = 1$: $\text{Tr}(T) = 1 \cdot 2 = 2$. *This is the first appearance of the “2” — it is the trace, predicted before any matrix is written down, from the number-operator structure of T2.*

Structural input 3: Rail-flip $\mathbb{Z}_2$ acts on V ’s image $\Rightarrow$ rank 1. By construction $T = V + PVP$ where P is the rail-flip ($P|(m_1, m_2)\rangle = |(m_2, m_1)\rangle$, $P^2 = I$). So $[P, T] = 0$, and T diagonalises in the ± 1 eigenspaces of P (symmetric / antisymmetric rail modes).

T2 and T4 (the two non-vanishing terms on the single-particle sector) both demand $m_1 \geq 1$, so V annihilates the no-rail-1-particle state $|(0,1)\rangle$. Equivalently, the image of V on $\mathcal{B}$ is one-dimensional. The same logic applied to $\tilde{V}$ gives $\text{im}(\tilde{V})|_{\mathcal{B}}$ also one-dimensional. Both images live in the same antisymmetric 1-d subspace — because the unique non-trivial rail-asymmetry of T4 (the $-n_2$ sign) is exactly what flips the output to the other rail. Hence $\text{rank}(T) = 1$, $\det(T) = 0$.

Conclusion (the eigenvalue 2 is predicted). A 2×2 matrix with $\text{Tr}(T) = 2$ and $\det(T) = 0$ has eigenvalues $\{0, 2\}$. Both numbers came from structural inputs:

- $\text{Tr}(T) = 2$ is the number-operator structure of T2 evaluated on the single-particle sector.
- $\det(T) = 0$ is the rail-flip $\mathbb{Z}_2$ acting on V 's rank-1 image.

The dominant eigenvalue is $\lambda = \text{Tr}(T) = 2$, on the antisymmetric rail mode. Iterating the chain, T^n scales the dominant mode by $\lambda^n = 2^n$. The leftmost vertex projects the boundary $|(1,0)\rangle$ onto exactly that mode (because V 's image is precisely the antisymmetric 1-d subspace, by Input 3). So the n -bracket ladder evaluates to 2^n *by structure, not by computation*.

Constructing $\hat{V}(z, w)$ explicitly — and watching it collapse. Gunnar's email asked specifically “what is $T(z)$ and how does Claude calculate it?” — so we should not just *assert* that the four variables drop out. Let us do the calculation.

Encode each leg occupation by a marker: input occupations (m_1, m_2) by $z_1^{m_1} z_2^{m_2}$, output occupations (n_1, n_2) by $w_1^{n_1} w_2^{n_2}$. The four-term coupling becomes the formal polynomial

$$\hat{V}(z_1, z_2; w_1, w_2) = \sum_{m_1, m_2, n_1, n_2 \geq 0} V(m_1, m_2; n_1, n_2) z_1^{m_1} z_2^{m_2} w_1^{n_1} w_2^{n_2}.$$

This is the object Gunnar said “depends on four variables, so it's a mess.” He is correct: as written it is a polynomial of unbounded degree.

Now project. The boundary $|(1,0)\rangle$ at both ends restricts both input and output to total-particle-number 1. In the polynomial, this means we keep only the bidegree-(1,1) part: total degree 1 in $\{z_1, z_2\}$ and total degree 1 in $\{w_1, w_2\}$. Walking the four terms:

- T1 has $m_1 = n_1 + 2$, so input has degree ≥ 2 in z_1 . **Killed by projection.**
- T2 has $m_1 = n_1$, $m_2 = n_2$, weight n_1 . Bidegree-(1,1) contribution: $(m_1, m_2, n_1, n_2) = (1, 0, 1, 0)$ with weight 1, giving $+z_1 w_1$. Other diagonals at total particle 1 have $n_1 = 0$ (weight 0). Net: $+z_1 w_1$.
- T3 has $m_2 = n_2 + 1$ and $m_1 = n_1 + 2$; bidegree-(1,1) would need $m_1 + m_2 = 1$, i.e. $n_1 + n_2 \leq -2$. **Killed.**
- T4 has $m_1 = n_1 + 1$, $m_2 = n_2 - 1$, weight $-n_2$. Bidegree-(1,1) contribution: $(m_1, m_2, n_1, n_2) = (1, 0, 0, 1)$ with weight -1 , giving $-z_1 w_2$.

So the projection of $\hat{V}$ onto the bidegree-(1,1) subspace is

$$\hat{V}|_{(1,1)}(z, w) = z_1 w_1 - z_1 w_2.$$

Two terms. *The other infinitely many terms of $\hat{V}$ exist but contribute zero to this observable, by the ΔN argument of Input 1.*

Add the rail-flip. $\tilde{V}$ is $\hat{V}$ with $z_1 \leftrightarrow z_2$ and $w_1 \leftrightarrow w_2$:

$$\tilde{V}|_{(1,1)}(z, w) = z_2 w_2 - z_2 w_1.$$

Sum:

$$\boxed{\hat{T}(z, w) \equiv \hat{V}|_{(1,1)} + \tilde{V}|_{(1,1)} = z_1 w_1 - z_1 w_2 - z_2 w_1 + z_2 w_2 = (z_1 - z_2)(w_1 - w_2).}$$

Reading off the matrix and the eigenvalue. The polynomial $\hat{T}(z, w) = (z_1 - z_2)(w_1 - w_2)$ *factorises*. That factorisation is the answer to “what is $T(z)$ ”:

- Reading $\hat{T}$ as a 2×2 matrix in the basis $\{z_1, z_2\} \otimes \{w_1, w_2\}$, the four coefficients of $z_i w_j$ are the four entries: $T_{11}=1, T_{12}=-1, T_{21}=-1, T_{22}=1$. There is no z -dependence remaining — the z_i, w_j have done their job as bookkeeping markers and are now read as basis labels.
- The factorisation $(z_1 - z_2)(w_1 - w_2)$ is rank-1 *by inspection*: input mode $z_1 - z_2$ (antisymmetric) couples to output mode $w_1 - w_2$ (antisymmetric) and to nothing else. Substituting $z_1=1, z_2=-1$ (the antisymmetric input) into $\hat{T}$ gives $(1 - (-1))(w_1 - w_2) = 2(w_1 - w_2)$. **Eigenvalue 2 on the antisymmetric mode, in one line, with no matrix arithmetic.** Symmetric input $z_1 + z_2$ gives $\hat{T}(1, 1; w) = 0$: kernel.

That is what we mean by “the four variables drop out.” The four-variable polynomial $\hat{V}$ is the right starting point. The bidegree projection (Input 1) eliminates the unbounded-degree pieces. The factorisation of the projected polynomial $(z_1 - z_2)(w_1 - w_2)$ exhibits the eigenvalue and rank in closed form. None of the steps required computing T^n or solving a characteristic polynomial.

Numerical cross-check. Reading off the matrix from $\hat{T}$, $T = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. Brute-force occupation sum agrees with $T^n V|(1, 0)\rangle = 2^n(|(1, 0)\rangle - |(0, 1)\rangle)$ through $n = 5$ (`verify_2x2.py`, in this letter directory — runs the polynomial collapse symbolically and the brute-force sum numerically).

Notation note. $\tilde{V}$ is the rail-flip $((m_1, m_2) \leftrightarrow (m_2, m_1))$, *not* matrix-transpose. Our previous draft wrote $V^\top$ for the rail-flip; that was the ambiguity Gunnar’s email was reacting to.

Where this fits in the AC programme. Same template as Lagrange inversion on $G(z) = z(1+G^2)$ for Gribov tree counts ($[z^3]G = 1, [z^5]G = 2, [z^7]G = 5$): vertex polynomial $\rightarrow$ boundary-projected sector $\rightarrow$ closed-form residue / eigenvalue. R3 below shows the same template runs end-to-end through the JT05 Eq.(60) exponents.

R3. End-to-end RG for Gribov: what AC delivers (answering his Q2)

Gunnar’s second question, paraphrased: “*the small handful of diagrams I look at are infinitely many for Claude. Can AC reduce that infinite set to a small handful, and does it apply to φ^4 in Doi–Peliti?*”

The short answer: **yes, and we have already done it for Gribov / DP at 2 loops.** The full calculation is in `paper/cfac/gribov.tex` (the “Gribov through the CFAC lens” draft); the verification suite is `rdft/ac/gribov/run_all_tests.py`. We summarise here what is delivered, what is input, and what would carry over verbatim to φ^4 in Doi–Peliti.

The reference target. JT05 (Janssen & Täuber, *J. Stat. Phys.* 2005, arXiv:cond-mat/0409670 — the canonical 2-loop DP paper). Specifically:

JT05 Eq. (57)	2-loop Z -factors $\{Z_\psi, Z_\lambda, Z_\tau, Z_u\}$	ε -Laurent series
JT05 Eq. (58)	RG functions $\gamma, \zeta, \kappa, \beta$	to $O(u^2)$
JT05 Eq. (59)	Wilson–Fisher fixed point u^*	to $O(\varepsilon^2)$
JT05 Eq. (60)	Critical exponents $\eta, z, \nu, \beta_{\text{DP}}$	to $O(\varepsilon^2)$

Täuber’s 2014 textbook (*Critical Dynamics*, CUP) contributes the standard relation $\gamma_X(u) = -u \partial_u$ (simple-pole residue of Z_X) used to go from Z -factors to RG functions; that is the only role the textbook plays.

Pipeline (counting layer $\rightarrow$ exponents).

$$\underbrace{G(z) = z(1+G^2)}_{\text{C1: Lagrange}} \rightarrow \underbrace{\sigma_\Gamma}_{\text{C2: bivariate}} \rightarrow \underbrace{U_\Gamma}_{\text{C3: Kirchhoff}} \rightarrow \underbrace{a_\bullet^X, \beta_1}_{\text{C4–C7: algebra}} \rightarrow \underbrace{Z_X^{(2,2)}}_{\text{C5: Hopf}} \rightarrow \underbrace{\{B_2, B_3^{\text{sun}}, B_V\}}_{\text{C6/C8: IBP}} \rightarrow \underbrace{Z_X^{(2,1)}}_{\text{assembly}} \rightarrow \dots$$

Honest scorecard against JT05.

JT05 quantity	CFAC status	Where
Diagram counts c_Γ	derived (Lagrange inversion)	<code>gribov.tex</code> §
Symmetry factors, signs σ_Γ	derived (Lagrange + bivariate marking)	<code>gribov.tex</code> §
Symanzik polynomials U_Γ	derived (Kirchhoff)	<code>gribov.tex</code> §
1-loop algebra $\{1/4, 1/8, 1/2, 2\}$, $\beta_1 = 3/2$	derived	<code>gribov.tex</code> §
Eq. (57) double poles	derived (Hopf antipode, closed-form rational)	<code>actrick.py</code>
Eq. (57) simple poles (12 IBP rationals)	derived (CFAC closure)	<code>ibp_coeffi</code>
Eq. (58) γ, ζ, κ	derived (Täuber relation on Z-factors, this letter)	<code>eq58_from</code>
Eq. (58) $\beta(u)$ at 1-loop	derived ($\beta_1 = a_u^{(1)} - 2a_{qb}^{(1)} = 3/2$)	<code>eq58_from</code>
Eq. (58) $\beta(u)$ at 2-loop	pending (MSbar pole-closure algebra; not a missing theorem)	—
Eq. (59) fixed point u^*	derived (series solution, zero residual)	<code>two_loop.p</code>
Eq. (60) exponents η, z, ν, β_{DP}	derived (zero residual vs JT05)	<code>two_loop.p</code>
3 master integral <i>values</i>	input (JT05 / Panzer / Borinsky)	<code>gribov.tex</code> §

What is genuinely input. *Two* items remain:

1. **Master integral values** $B_2, B_3^{\text{sun}}, B_V$. CFAC reduces 8–10 distinct loop integrals to these three (via IBP) but does not produce their numerical values; those are quoted from JT05 / Panzer / Borinsky. AC has nothing to say here — this is where *integration* enters and *counting* stops, by design.
2. **2-loop coefficient of $\beta(u)$** . The 1-loop part $\beta_1 = 3/2$ is CFAC-derived; the 2-loop coefficient $-169/128 - 106L/128$ requires the standard MSbar 2-loop pole-cancellation closure (Vasil’ev Eq. 1.114), which we have not yet written out symbolically. Once written, it follows from the same Z -factor data the γ, ζ, κ derivations use — bookkeeping, not a missing theorem. With $\beta(u)$ from JT05 as a placeholder, the SymPy pipeline produces JT05 Eq. (60) with *zero residual*.

Everything else — counts, signs, symmetry factors, double poles, simple-pole rationals, three of four RG functions, fixed point, exponents — is CFAC closed-form, with zero hand-enumerated diagrams.

The reduction Gunnar asked about (“infinite to handful”). The entire 2-loop DP topology list is $[z^3]G + [z^5]G + [z^7]G = 1 + 2 + 5 = 8$ graphs, all delivered by one polynomial equation $G(z) = z(1 + G^2)$ via Lagrange inversion. IBP reduces 8–10 loop integrals to *three* masters; two of those are textbook. All four 2-loop double poles are closed-form rational (Hopf antipode), no integration. So the “infinite set” compresses to: $G(z) = z(1 + G^2)$ for counts, three master integral values, Hopf antipode for double poles.

φ^4 in Doi–Peliti. Same template with $\phi(G) = 1 + G^4$ in place of $1 + G^2$: rerun Lagrange inversion, redo the algebra on the new vertex pair, redo Hopf antipode. We have not run this case end-to-end — happy to do so on receipt of Gunnar’s preferred action / vertex normalisation.

A The actual calculations

A.1 A. φ^4 Wick weights as generating-function coefficients

Take the labelled-source generating function

$$\mathcal{Z}(g, r; J) = \int \mathcal{D}\varphi \exp\left\{ -\frac{1}{2} \int (\nabla\varphi)^2 - \frac{r}{2} \int \varphi^2 - g \int \varphi^4 + \int J\varphi \right\}.$$

Expanding in g produces the labelled-graph sum with Wick weights:

$$\mathcal{Z}(g, r; J) = \mathcal{Z}_0[J] \sum_{k \geq 0} \frac{(-g)^k}{k!} \left\langle \left(\int \varphi^4 \right)^k \right\rangle_{0, J}.$$

Tadpole (g^1 , two-point function). The one-loop self-energy is the g^1 -coefficient of $-\partial_{J(x)} \partial_{J(y)} \log \mathcal{Z}$ at $J = 0$:

$$[g^1] \Sigma(x, y) = - \langle \varphi(x) \varphi(y) \int dz \varphi(z)^4 \rangle_{0, \text{conn}} = - 12 \int dz G(x-z) G(0) G(z-y).$$

The $12=4 \cdot 3$ falls out of the Gaussian moment as Wick pairings: 4 ways for $\varphi(x)$ to land on a z -leg, 3 for $\varphi(y)$ on a remaining z -leg, the last two z -legs make the loop. *Identical to Gunnar's $4 \cdot 3$ in §1(a)*, read off as a Gaussian-moment derivative.

Bubble (g^2 , four-point function).

$$[g^2] \Gamma^{(4)} = \frac{1}{2!} \int dz_1 dz_2 \langle \varphi^4(z_1) \varphi^4(z_2) \rangle_{1\text{PI}, 4\text{-ext}} = \binom{4}{2}^2 \cdot 2 \cdot 4! \cdot \frac{1}{2} \int \frac{d^d k}{(k^2 + r)^2}.$$

Decomposes as Gunnar wrote it: $\binom{4}{2}^2$ for loop-leg choices, 2 for matching internal lines, 4! for external slot assignments, $\frac{1}{2}$ for vertex permutation. *Identical to Gunnar's §1(b)*.

Resulting exponent. Feeding these into β_u and Z_r at $g^*=\varepsilon/36$ gives $\nu = \frac{1}{2} + \frac{\varepsilon}{12}$ on the nose, by his own algebra.

A.2 B. Verifying the 2×2 reduction explicitly

We numerically confirmed everything in R2 against a brute-force sum over occupations. The verification script is bundled in this letter directory and reproduces:

- V matrix on $\mathcal{B}$: $\begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}$; $\tilde{V}$ matrix on $\mathcal{B}$: $\begin{pmatrix} 0 & -1 \\ 0 & -1 \end{pmatrix}$; $T = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$; $T^2 = 2T$.
- The chain element $\langle (1,0) | T^n V | (1,0) \rangle = 2^n$ for $n = 0, \dots, 5$.
- Brute-force $\sum_{\ell_1, \dots, \ell_{2n}}$ on the four-term Kronecker structure (truncated at $\ell_i \leq 4$, more than enough by Step 1) gives 2, 4, 8, 16, 32 for $n = 1, \dots, 5$, matching.

The point of the matrix derivation in R2 is that none of this needed Mathematica or even brute-force enumeration: each rung's factor of 2 is the eigenvalue of T on the boundary mode; the chain factorises completely.

Why the boundary sector is closed (restated). V has terms with $\Delta N = 2, 0, 3, 0$, so it is not particle-conserving. But on $| (1,0) \rangle$ and $| (0,1) \rangle$, the $\Delta N \neq 0$ terms require negative occupations on output and so vanish identically. The dynamics close on $\mathcal{B}$, by the structure of the four-term coupling, on this particular sector only.

Why the z 's drop out. The full operator $\hat{V}(z_1, z_2; w_1, w_2)$ contains pieces of every degree (e.g. the $\delta_{m_2, n_2+1} \delta_{m_1-1, n_1+1}$ term sits at total in-degree ≥ 3). The boundary $| (1,0) \rangle$ projects onto in-degree 1, in-degree 1 states only contract through in-degree-1 vertices, and the in-degree-1 truncation of $\hat{V}$ has only four nonzero coefficients — which assemble into the 2×2 matrix V above with no remaining z -dependence.

What to send back

1. **Q1 (the 2×2).** R2 derives the eigenvalue 2 from three structural inputs (boundary selects $N=1$ sector of dim 2; Term 2 is the rail-1 number operator $\Rightarrow \text{Tr}(T) = 2$; V annihilates rail-1-empty states $\Rightarrow \det(T) = 0$). The polynomial route exhibits this in one line: $\widehat{T}(z, w) = (z_1 - z_2)(w_1 - w_2)$, rank-1 by inspection, eigenvalue 2 on the antisymmetric mode. *Verified through $n = 5$ by brute-force* (`verify_2x2.py`).
2. **Q2 (Gribov RG end-to-end).** R3 closes the pipeline: $G(z) = z(1 + G^2)$ for counts $\rightarrow$ Hopf antipode for all double poles $\rightarrow$ IBP closure for simple poles $\rightarrow$ Tauber relation for γ, ζ, κ (this letter, `eq58_from_z.py`) $\rightarrow$ SymPy series for u^* and exponents $\rightarrow$ JT05 Eq. (60), *zero residual* (`rdft/ac/gribov/run_all_tests.py`). Two items remain input: 3 master integral values, and the 2-loop $\beta(u)$ coefficient pending MSbar pole-closure algebra. Template extends to φ^4 -in-Doi-Peliti by $\phi(G) = 1 + G^2 \mapsto 1 + G^4$.
3. **The connection.** The 2^n of his §2 plays the same role in his theory that $4 \cdot 3$ plays in φ^4 : a per-diagram counting factor that feeds the RG downstream. One template handles both — vertex polynomial $\rightarrow$ boundary-projected sector $\rightarrow$ trace / eigenvalue / Lagrange residue — and runs all the way to critical exponents for Gribov.